Mine Diversified Contents of Multispectral Cloud Images Along With Geographical Information for Multilabel Classification

Dongxiaoyuan Zhao, Qiong Wang, Jinglin Zhang, and Cong Bai

***Abstract*— Multispectral multilabel cloud image classification (MSMLCIC) aims to predict a set of labels presented in a multispectral (MS) cloud image, which usually contains more than one cloud type or weather system. However, the exploration of diversified contents reflected by multiple bands of MS image is limited and the consideration of geographical information (time and location information) is insufficient. To cope with the abovementioned problems, this work proposes the multispectral cloud image multilabel classifier with group feature extractor and geo-queries (MS-GoGo). With a group feature extractor, different bands of MS images are processed separately according to the content they reflected, and a group of distinctive yet complementary image features are generated. Geo-queries are responsible for implicitly embedding different labels with time and location information to probe the corresponding similar semantic ingredients. Due to the coarse classification of the existing dataset, a new dataset named LSCIDMR-V2 is generated with fine-grained cloud-type annotation and multichannel data. The experiment shows that, using the group feature extractor and geo-queries, the popular used metric subset accuracy is improved from 40.06 to 42.87 and 44.35, respectively. The proposed method achieves the mean average precision of 82.40, outperforming state-of-the-art methods.**

***Index Terms*— Cloud type, deep learning, multilabel classification, multispectral (MS), remote sensing (RS), weather system.**

I. Introduction

WEATHER condition plays an important role in our daily life. Clouds, being an extremely influential part of various weather systems, have long been a focus of observation and study in the meteorological research community. Advanced observation systems, constantly evolving theories, and unprecedented computing power give us powerful tools to identify, understand, and predict clouds as well as weather systems better than ever. In the context of different scenarios such as tropical cyclone (TC) estimation [1], [2], [3], rainfall prediction [4], [5], [6], data correction [7], [8], change detection [9], [10], [11], cloud detection [12], [13], [14], and so on, deep-learning methods have already been widely applied in meteorological research. Meanwhile, with the help of deep learning, using computers to process and understand images becomes a common practice. Deep-learning-based methods could also be great tools for weather system recognition and prediction by understanding satellite cloud images. Also, this problem can be seen as remote sensing (RS) image classification.

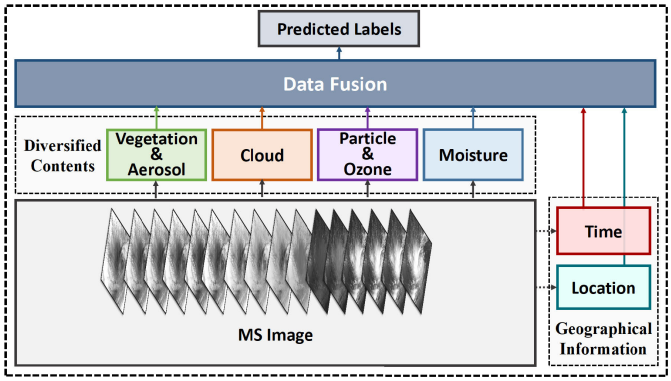


Fig. 1. Motivation of MS-GoGo. Various contents (vegetation, aerosol, cloud, particle, ozone, and moisture) reflected by MS images need more distinctive representations. Geographical information (time and location information) should be introduced into MS-oriented models since they are closely related to the formation and development of clouds and weather systems.

Most of the methods developed for RS image classification are single-labeled [15], [16], [17], which means that only the most significant label would be learned by the model. However, a real-world RS image always contains more than one object that can be annotated into different categories, and thus, multilabel image classification (MLIC) has been introduced in RS. A model can not only learn supervised label information but also the relationship between different labels from multilabeled RS images. There are also a number of works on remote sensing MLIC (RSMLIC), adopting general computer vision (CV) tools to deal with RS problems. However, to solve RSMLIC problems, more distinctive characters of RS data should be considered and the rich information of it requires further exploration. Taking multispectral (MS) images for example, as shown in Fig. 1, various contents reflected by MS images need more distinctive representations and the introduction of valuable time and location information

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is required. For the purposes of taking full utilization of the diversified information contained in the MS image as well as the geographic information it carries to improve the RSMLIC performance, this work proposes a geographical information considered and MS information well-organized RSMLIC model named multispectral cloud image multilabel classifier with group feature extractor and geo-queries (MS-GoGo). MS-GoGo can take full advantage of MS images' information, more than simply see the superficial visual signals in MS images. The source code of MS-GoGo will be released.[1]

Another challenge to overcome is the demand for BIG DATA. As well known, data are the "fuel" of deep learning. A well-trained deep-learning model, to a great extent, comes from the support of a well-annotated and large-scale dataset. Unfortunately, RS benchmarks were limited in scale and amount until 2019. Furthermore, because of the domain gap, models pretrained on large-scale general CV archives such as ImageNet [18] are not applicable in satellite cloud image tasks. To overcome the limitations of data scale, Sumbul et al. [19] brought out an available large-scale RS benchmark named BigEarthNet, which is annotated with image-level multilabels. BigEarthNet pioneers the way in large-scale RS image archives, but as a land-cover-used dataset, for the application related to weather systems and cloud types, it (even upgraded BigEarthNet-MM [20]) still could not meet the requirement. Thus, to fill such a data gap, Large-scale Satellite Cloud Image Database for Meteorological Research (LSCIDMR) [21] proposed a weather and cloud-oriented dataset, which is annotated with both image-level single labels and multilabels. However, cloud, as a significant part of weather systems, was coarsely classified into two cloud phases (high ice cloud and low water cloud) in LSCIDMR. In consideration of the different impacts of different clouds on weather systems, we need a more fine-grained classification of cloud types. Moreover, LSCIDMR only used three channels of Himawari-8 satellite sensors that are mostly related to cloud and weather systems, but we think that it is better if we also use other channels that contain rich information about clouds and atmospheric composition. For these two reasons, the LSCIDMR is extended in two respects in this work: labels of two cloud phases will be replaced with nine cloud types and all 16 channels of Himawari-8 satellite sensors will be used to construct LSCIDMR-V2.

The main contributions of this article can be summarized as follows.

1) Group features extractor is proposed for the better utilization of diversified yet complementary information contained by MS cloud images separately since it can handle different contents of MS images and extract more distinctive high-level features of them.
2) Geo-queries is proposed, which can effectively utilize geographic information and introduce such information into deep-learning models.
3) A new dataset named LSCIDMR-V2 for the use of weather systems, cloud types, and land-cover classification is extended from the original LSCIDMR [21]. LSCIDMR-V2 has more fine-grained cloud-type annotation and multichannel data.
4) The effective utilization and fusion of various information improved the model's performance on RSMLIC.

[1]Github link: https://github.com/Zjut-MultimediaPlus/MS-GoGo

II. Related Work

A. Multilabel Image Classification

As the basis of RSMLIC task, some ideas and methods of general MLIC can also be used in RSMLIC. In this section, recent progress in MLIC will be briefly reviewed.

As reviewed in [22], single-label classification models can be used to do MLIC tasks by regarding it as a set of independent binary classification problems. However, this approach ignores semantic and spatial correlations among different labels, which prevents the performance of high accuracy. Sequence-to-sequence (Seq2Seq) methods and graph-based methods are two kinds of mainstream approaches to take advantage of the semantic relationship between labels. As a kind of Seq2Seq model, recurrent neural network (RNN) is first used by the convolutional neural network (CNN)–RNN unified framework [23] in 2016 to model label dependencies in MLIC. The proposal of C2AE [24] provided a paradigm of MLIC, in which models first projected the image features and corresponding semantic label representations into the same latent space to get a joint feature and label embedding, and then, the embedding would be used to recover the label. Later works, such as [25], [26], [27], and [28], frequently use Seq2Seq models, such as RNNs, long short-term memory (LSTMs), and transformers [29] structures to model the relationships among different labels. Utilizing graph-based models, graph convolutional network (GCN) [30] is another way to extract such correlations since the graph is naturally suitable for the illustration of relationships. Chen et al. [31] and Wang et al. [32] used a statistical approach to build a relationship graph for MLIC. To solve the problem of data distribution generalization and to get rid of the tediousness of manual graph construction, Ye et al. [33] proposed ADD-GCN to build graphs dynamically. How to utilize the spatial relationship of labels is also explored. Zhang et al. [25] used an RPN-like [34] fully convolutional localization layer to localize regions that potentially contain multiple labels. Zhu et al. [35] used a spatial regularization net to get label attention maps, which exploits both semantic and spatial relations between labels. Attention mechanism was also used in [36], in which cross-modal attention was proposed to capture spatial and semantic relationships among labels. A multiscaled transformer unit was used to capture the long-range structural relation of different objects in the image, and along with semantic relation, a joint relationship graph was built in [37]. To a better localization of interested regions, residual attention was proposed in [38], in which the main features are weighted by the average pooling attention map, while the residual features are weighted by the max-pooling attention map.

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B. Remote Sensing Multilabel Image Classification

Although the abovementioned MLIC methods could be transplanted into the RS field, considering the specificity of the RS data, some RS-oriented MLIC models have been brought forward.

For example, Bazi [39] used a two-branch network to deal with the image and labels. Hua et al. [40] extracted different features from different labels and inferred relationships between feature pairs. Hua et al. [41] used classwise attention maps to focus on different object regions, and a bidirectional LSTM was used in consideration of bidirectional class dependencies. Considering that RS image bands are not completely consistent in spatial resolution, Sumbul and Demir [42] proposed a K-branch CNN to obtain local descriptors and multiattention was used to obtain local descriptors' attention scores. Diao et al. [43] adopted deformable convolutions to adaptively extract features with geometric transformation invariance and adaptive receptive field, which fit the characters of the RS image. Li et al. [44] employed a superpixel segmentation algorithm to get a set of nonoverlapping regions, which could be considered as vertices of a graph, and together with image features, a graph convolution operation was applied to map the region adjacency graph to multilabel prediction vectors. CEGCN [45] uses CNN and GCN to capture complementary features of small-scale regular regions and large-scale irregular regions, respectively, for better pixel-level hyperspectral image classification performance. Considering the noise contained in multilabel images' annotation, Burgert et al. [46] brought forward a noise-adaptive training process. In view of the expensive process of multilabel annotation, Yu et al. [47] proposed a weakly supervised model to conduct multilabel aerial image classification.

Although the general multilabel classification method could be directly applied in the RS field and some approaches customized for RS images have been brought forward, there is still room for further exploration of the contents' diversity of different bands and the effective utilization of geographic information, which we try to cope with in this article with the proposed MS-GoGo.

C. RS Benchmarks

Different RS benchmarks were brought forward for different application scenarios, these datasets were collected from different data sources (data platforms such as Google Earth or satellites such as Sentinel) with different annotation types (single-labeled or multilabeled). This section will briefly introduce RS benchmarks that we have known.

The scale of RS datasets experienced the development from small to large. Original UC-Merced [48], a 1000-level dataset, was proposed in 2010 with single-labeled annotation and it was extended to a multilabel form [49] eight years later. In the years that followed, Google Earth became a popular data source, on which a series of single-label land-use datasets (WHU-RS19 [50], NWPU-RESISC45 [54], and SIRI-WHU [52]) were proposed, while the scale is gradually grown to 10 000 levels. BigEarthNet [19] is the latest 100–1000-level large-scale land-cover dataset with multilabel annotations referring to the CORINE Land Cover database of the year 2018 (CLC 2018[2]). BigEarthNet-MM [20] is a multimodal land-cover benchmark extended from BigEarthNet.

Datasets mentioned above are mostly focused on land-use and land-cover scene classification, and there is no cloud image classification benchmark until LSCIDMR [21]. LSCIDMR images are synthesized from red bands and two near-infrared channels since these three channels are most representative of the categories labeled in LSCIDMR [21]. However, as a significant role in the weather system, clouds were coarsely divided into two cloud phases (high ice cloud and low water cloud), and the 16-channel data provided by Himawari-8 can be further explored. Because of these two reasons, this article extends LSCIDMR from two cloud phases to nine cloud types, and all 16 bands from Himawari-8 are included in the new dataset. The extended dataset is named LSCIDMR-V2.

The benchmarks we know are listed in Table I for convenience of reference.

III. Method

In this section, we will describe the overall architecture of the proposed MS-GoGo and the design of its components in detail.

A. Framework

Given an input RS image x with time and location information g and its corresponding labels $y = [y_1, y_2, \ldots, y_C]$, where $y_i \in \{0, 1\}, i = 1, \ldots, C$, in which the value 1 and 0 of y_i means positive or negative of ith label's existence, respectively, the target of our goal is to find a mapping from x (contains g) to y, and the prediction is denoted by $\hat{y} = [\hat{y_1}, \hat{y_2}, \ldots, \hat{y_C}]$, where $\hat{y_i} \in [0, 1], i = 1, \ldots, C$ and the value of $\hat{y_1}$ shows the probability of the ith label's appearance in x.

Fig. 2 shows the overall architecture of the proposed MS-GoGo. The left part is the design of group feature extractor, which is responsible for the separate processing of different content features. Groups of bands of input RS images' features are extracted from a set of independent CNNs. These features of diversified contents are then sent to a transformer encoder block to perform data fusion processing. Content querying is performed in conjunction with data fusion, in which time and location information and label dependencies are introduced to our model. After the data fusion and content querying, all information relating to the existence of labels is pooled to the output sequence ($\mathbf{p}$) of transformer encoder block, and it is then fed into a feedforward network for the purpose of predicting the appearance of the queried labels.

B. Group Feature Extractor

Compared with RGB images, MS satellite images contain more abundant and diversified information since different bands reflect various contents of Earth surface. In view of this, how to effectively mine the richness and take the advantage

[2]Website: https://land.copernicus.eu/pan-european/corine-land-cover

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TABLE I
COMPARISON BETWEEN THE PROPOSED LSCIDMR AND OTHER PUBLICLY AVAILABLE SATELLITE IMAGE DATABASES

Dataset Name	Data Sources	Annotation Type	Image Type	Total Images	Classes	Scenario	Year
UC-Merced[48], [49]	USGS*	Single Multi.	Aerial RGB	2100	21 17	Land-use	2010 2017
WHU-RS19[50]	Google Earth	Single	Aerial RGB	1005	19	Land-use	2012
RSSCN7[51]	Google Earth	Single	Aerial RGB	2800	7	Land-use	2015
SIRI-WHU[52]	Google Earth	Single	Aerial RGB	2400	12	Land-use	2016
RSCII[53]	Google Earth	Single	Aerial RGB	1232	11	Land-use	2016
NWPU-RESISC45[54]	Google Earth	Single	Aerial RGB	31500	45	Land-use	2016
AID[55]	Google Earth	Single	Aerial RGB	10000	30	Land-use	2017
Brazilian Coffee Scene[56]	SPOT	Single	Satellite Red, Green & NIF◇	2876	4	Agricultural	2015
EuroSAT[57]	Sentinel-2	Single	Satellite RGB & MS†	27000	10	Land-use	2015
PatternNet[58]	Google Earth	Single	Aerial RGB	30400	30	Land-use	2018
BigEarthNet[19]	Sentinel-2	Multi.	Satellite MS	590326	43	Land-cover	2019
BigEarthNet-MM[20]	Sentinel-1&2	Multi.	Satellite MS & RS Radar	590326	43	Land-cover	2021
LSCIDMR[21]	Himawari-8	Single & Multi.	Satellite Red & NIF	104390	10	Clouds & Weather	2020
LSCIDMR-V2	Himawari-8	Single & Multi.	Satellite MS	104390	17	Clouds & Weather	2022

* Website: https://www.usgs.gov/
◇ NIF: Near-Infared
† MS: Multi-Spectral

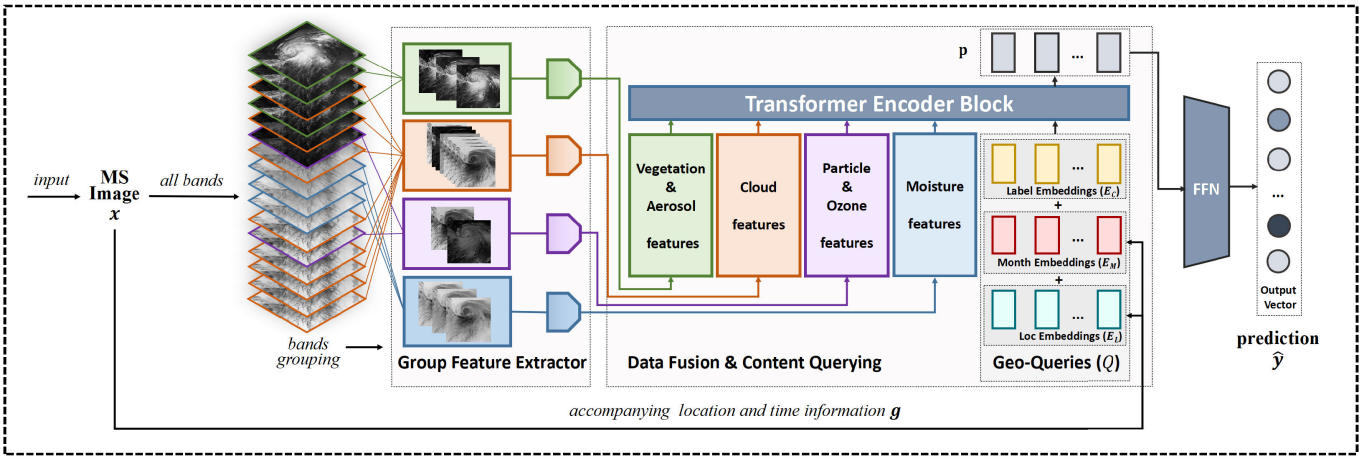


Fig. 2. Overview of the proposed MS-GoGo. Channels of input MS image are first grouped according to the contents they reflect (represented by different colors), and each group is handled by an independent CNN to get the corresponding high-level features. Geo-queries (Q) composed of label embeddings (E_C), month embeddings (E_M), and loc embeddings (E_L), along with features extracted by the CNNs, are fed into a vanilla transformer encoder to perform data fusion and content querying. The output sequence ($\mathbf{p}$) of the transformer encoder block is then processed by a feedforward network to get the output likelihood of each label's existence.

of the diversity of information that MS images hold should be well considered.

To extract the features of MS images, we can simply treat MS images as a whole and apply a single CNN or other backbones to do the feature extraction. However, it is not wise way to just pour the data into a model and let the model to decide what to learn by itself. Also, such a way could be seen as an early fusion of different modalities (contents), and as pointed out in [59], a model with a feature fusion operation in the early stage of the network may not be able to get a promising result because premature fusion can lead to low efficiency of high-level representation of particular contents. There is a classical architecture in the video understanding field, named two-stream convolutional network [60], which considered the information from frames and objects' movement tendency separately and achieved tremendous success on multiple tasks. Inspired by that, in this article, we propose a group feature extractor, which handles various contents reflected by different bands of MS images separately, to extract a group of distinctive yet complementary features for the later fusion.

To be more specific, as shown in the left of Fig. 2, different bands of an original MS image would be grouped first according to the contents they reflect (the physical properties of MS image from Himawari-8 are listed in Table II), and each group is concatenated and treated as a multichannel image. Also, a series of independent CNNs is trained to extract the features of these grouped bands. The kth feature $Feat_k$ extracted from the kth group of bands could be formalized as

$$\begin{aligned} \text{Feat}_k &= \text{Flat}(f_{\theta_k}([b_{k1}, b_{k2}, \ldots, b_{kn}]) \\ \text{Feat}_k &\in \mathbb{R}^{L\times D}, \quad L = W \times H \end{aligned} \tag{1}$$

where $f_{\theta_i}(\cdot)$ denotes the transform performed by the ith CNN, whose parameter is θ_i; W, H, and D are the widths, length, and depth of feature map generated by CNN, respectively; $Flat(\cdot)$ denotes the operation to flatten feature map into

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TABLE II
USAGE OF HIMAWARI-8'S 16 BANDS IN LSCIDMR AND LSCIDMR-V2 AND THEIR CORRESPONDING GROUPS IN MS-GOGO

Bands	Center Wave Length (μ m)	Physical Properties	Group	LSCIDMR	LSCIDMR V2
albeo_1	0.46	vegetation, aerosol	1	✗	✓
albeo_2	0.51	vegetation, aerosol	1	✗	✓
albeo_3	0.64	low cloud , fog	2	✓	✓
albeo_4	0.86	vegetation, aerosol	1	✓	✓
albeo_5	1.6	cloud phase	2	✓	✓
albeo_6	2.3	particle size	3	✗	✓
tbb_7	3.9	low cloud, fog , forest fire	2	✗	✓
tbb_8	6.2	mid and upper level moisture	4	✗	✓
tbb_9	7.0	mid level moisture	4	✗	✓
tbb_10	7.3	mid and upper level moisture	4	✗	✓
tbb_11	8.6	cloud phase , SO2	2	✗	✓
tbb_12	9.6	ozone content	3	✗	✓
tbb_13	10.4	cloud imagery	2	✗	✓
tbb_14	11.2	cloud imagery	2	✗	✓
tbb_15	12.3	cloud imagery	2	✗	✓
tbb_16	13.3	cloud top height	2	✗	✓

a vector sequence; $[\cdot]$ denotes the concat operation; and $b_{k1}, b_{k2}, \ldots, b_{kn}$ are n bands that belong to group k.

Consistent with what [19] pointed out, we also find in our experiments that shallow CNNs are more suitable for RS image feature extraction. Thus, in our implementation, every CNN we utilized consists of only three convolutional operations followed by a Relu as an activation function and a max pooling to do downsampling.

C. Geo-Queries

The location and time information of MS images are utilized by geo-queries along with each label's semantic information.

To take full advantage of label dependencies [23] (relation of different labels) in MLIC tasks, seq2seq methods regularly use embedding techniques to represent the semantic information of every single label, and the distance of labels' representation stands for the relationship among them. Specifically, in multihead self-attention (MSA)-based transformer architectures, the concept of object queries [61] and label queries [27] was brought forward to illustrate the detecting process of the existence of an object in images. In other words, queries are utilized to probe the corresponding similar semantic ingredients in image features and the relationship between different labels.

Nevertheless, queries, essentially being a sequence of learnable vectors, are flawed in dynamic relation representation. For example, in different seasons, the association between two labels varies a lot, which cannot be finely represented by a set of invariable vectors. In order to dynamically embed the semantic information of labels and introduce basic geographic information along with the label dependencies into our model, we propose geo-queries to implicitly embed different labels with time and location information. Geo-queries ($Q \in \mathbb{R}^{C\times D}$) consist of three components—label embeddings ($E_C \in \mathbb{R}^{C\times D}$), month embeddings ($E_M \in \mathbb{R}^{M\times D}$), and loc embeddings ($E_L \in \mathbb{R}^{P\times D}$), where C, red M, P, and D denote the number of categories, the number of months in a year, the number of positions, and the embedding size, respectively. Specifically, label embeddings are responsible for representing the semantic information of labels, while loc embeddings and month embeddings encode location and time information, respectively. Let us take a closer look at geo-queries, and if $q_i \in \mathbb{R}^D$ is the ith query of Q, it could be formalized as follows:

$$q_i = e_{Ci} + e_{Mt} + e_{Lp} \tag{2}$$

where e_{Ci} is the embedding of ith label and while e_{Mt} and e_{Lp} implicitly represent the month and location the RS image belongs to, respectively. The geo-queries can then be represented as

$$Q = [q_1, q_2, \ldots, q_i, \ldots, q_C]^\top. \tag{3}$$

A specific example of geo-queries with its corresponding RS image is shown in Fig. 3. In this example, the model learns the co-occurrence relationship and semantic representations of labels in August at location 08_14. Each geo-query is responsible for estimating the existence of its corresponding label. In the example of Fig. 3, the query of TC should be aware of the existence of its target information carried by image features. By adopting these three kinds of embeddings to form geo-queries, information of geographic location, time, and label dependencies are effectively included in our model.

D. Data Fusion and Content Querying With Transformer

In our previous pipeline, we have already represented diversified contents and geographic information in the form of a vector sequence. The next we need to do is to establish interactions of these kinds of information. Due to the capability of feature aggregating [62] and relation extracting capability [63], a transformer encoder is adopted to fuse various kinds of information in our design.

Transformer is originally utilized in natural language processing (NLP) tasks [29], [64]. Due to its effectiveness and efficiency, it has been gradually transplanted into CV tasks [65]. The core component of transformers is MSA. The formal representation of the self-attention mechanism is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q \times K^\top}{\sqrt{d_k}}\right) \times V \tag{4}$$

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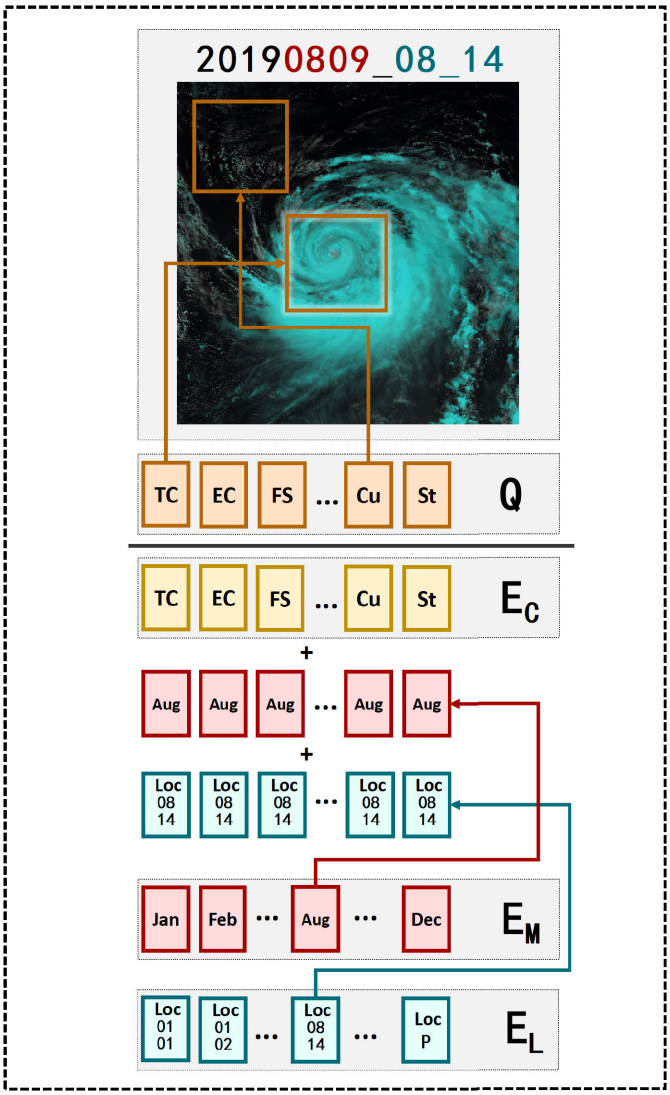


Fig. 3. Example of geo-queries. The corresponding geo-queries of RS image 20190809_08_14 is illustrated. The month embedding of August and the location embedding of loc_08_14 are selected from E_M and E_L, and then, they are summed with label embeddings to acquire geo-queries Q. Each geo-query is responsible for probing its interested ingredients from the feature of diversified contents.

where

$$Q = XW^Q \tag{5}$$

$$K = XW^K \tag{6}$$

$$V = XW^V \tag{7}$$

in which $X \in \mathbb{R}^{l \times d_k}$ is the input sequence (tokens), W^Q, W^K, and W^V are learnable matrices, and d_k is the number of feature dimension. From (4), we can see that, for one token (a vector from the input sequences), the self-attention mechanism allows it to focus on other tokens that it is most interested in, and hence, these tokens can be associated with each other. To take it a step further, the MSA tends to establish several distinctive relations among tokens to describe the associations more comprehensively. What is more, transformer is able to catch long-range dependencies of the input sequence since each pair of tokens has an attention value calculated.

The characteristics of transformers mentioned above make it very suitable for our purposes, which are looking for associations from different representations of various contents and helping queries to find features of interest. The tokenized vegetation and aerosol features, cloud features, particle and ozone features, as well as moisture features are concatenated along with the geo-queries and fed into the transformer encoder. Inside the MSA, the relationships of features are caught, and the associations between queries and their targets are established. After N layers of passing forward of the transformer encoder block, a sequence of vectors is outputted. The last class number of tokens $\mathbf{p} = [p_1, p_2, \ldots, p_C] \in \mathbb{R}^{C \times D}$ is selected as the final output of this module, following the practice of C-Tran [26].

E. Feedforward Network (FFN)

For the outputted sequence of the transformer encoder block, projection is needed for obtaining the probabilities of each label's existence. A linear projection is utilized followed by a Sigmoid function to limit the value within [0, 1]. Specifically, the probability of the ith label can be calculated as

$$\hat{y}_i = \text{Sigmoid}\left(\boldsymbol{\omega}_i^\top \cdot p_i + \boldsymbol{\beta}_i\right) \tag{8}$$

where $\boldsymbol{\omega_i} \in \mathbb{R}^{\mathbb{D}}$ and $\boldsymbol{\beta}_i \in \mathbb{R}$ are learned parameters of the linear projection, "$\cdot$" denotes the inner product operation, and $i = 1, 2, \ldots, C$. Notations $\boldsymbol{\omega}$ and $\boldsymbol{\beta}$ are defined as $\boldsymbol{\omega} = [\boldsymbol{\omega}_1, \boldsymbol{\omega}_2, \ldots, \boldsymbol{\omega}_C]^\top \in \mathbb{R}^{C \times D}$ and $\boldsymbol{\beta} = [\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \ldots, \boldsymbol{\beta}_C]^\top \in \mathbb{R}^C$, respectively. The prediction is finally notated as $\hat{y} = [\hat{y_1}, \hat{y_2}, \ldots, \hat{y_C}]$.

IV. Data Collection

To evaluate the effectiveness of our approach and provide support for future research, we extended the original LSCIDMR [21] to a form with MS data and fine-grained cloud types, named LSCIDMR-V2. In this section, we will briefly describe the extending process from LSCIDMR to LSCIDMR-V2 and basic information about LSCIDMR-V2.

The source data we used are provided by the Japan Aerospace Exploration Agency (JAXA) P-Tree System. Also, the grid data with a spatial resolution of 2.0 km and a temporal resolution of 10 min is selected. The time of source data we choose is UTC+0200 every day in 2019, and the northern hemisphere of the western pacific is covered in our dataset. The source data are cropped into 1000 pixel patches with a step of 200 pixels. The patch naming follows the format YYYYMMDD_ii_jj, where YYYY, MM, and DD denote the year, month, and day, respectively, and ii and jj indicate the location of a patch, which is the same as original LSCIDMR.

Different from the original LSCIDMR, all 16 channels are included in LSCIDMR-V2 since, in our perspective, the channels discarded in original LSCIDMR [21] are abundant in information about weather systems, clouds, and atmospheric composition. The detailed information of 16 channels is listed in Table II. What is more, different from the manual annotation, the extended labels in LSCIDMR-V2 are annotated automatically with the help of Advanced Himawari Imager (AHI) products. We adopt the CLoud Property (CLP) of the AHI

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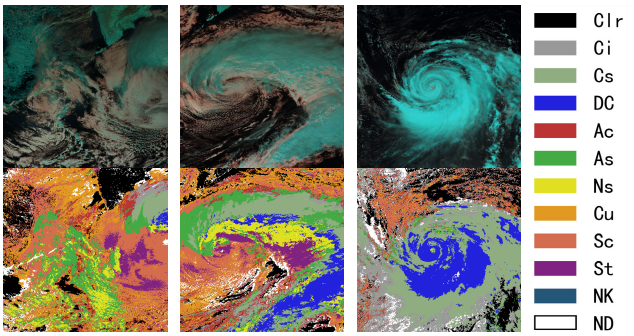


Fig. 4. Three examples of cloud type product. The first row shows synthetic images[4]of cloud patches, and the second row shows their corresponding cloud-type products' visualization. The legend on the right tells us the colors that stand for different cloud types, and Clr, NK, and ND denote clear in sky, not known, and no data, respectively. (a) Example 1. (b) Example 2. (c) Example 3.

L2 product as the guide of our annotation. The cloud types provided in CLP follow the ISCCP[3] international standard. Fig. 4 shows some examples of the cloud-type products in CLP with their corresponding satellite image patches.

The label-extending procedure could be summarized as follows.

1) *Single Label:*
   a) For every example which is labeled with high ice cloud, low water cloud, and labeless in the original LSCIDMR, we check the cloud type with the highest proportion to the sky coverage in this sample; if the proportion is higher than 20%, the patch would be labeled to that cloud type, and otherwise, this patch would be annotated with Labeless.
   b) For other examples, except TC, extratropical cyclone (EC), frontal surface (FS), and westerly jet (WJ), if the cloud coverage proportion is higher than 50%, we handle this patch as the same as step 1, and otherwise, nothing would be done to this patch.
2) *Multilabel:*
   a) The labels of the original high ice cloud and low water cloud are discarded.
   b) If the proportion of a cloud type of a sample in sky coverage is higher than 5%, the corresponding label would be attached to that sample.

After the above operation, the number of categories would be extended to 17. Also, the detailed information about the original LSCIDMR and LSCIDMR-v2 could be summarized in Table III.

V. Experiments

To evaluate the performance of our proposed MS-GoGo, we conduct experiments on LSCIDMR-V2 to compare our model with other popular methods. To demonstrate the role of each component in our model, a series of ablation experiments is presented after an overall comparison.

[3]Website: https://isccp.giss.nasa.gov/cloudtypes.html

[4]synthesized from 3rd, 4th, and 5th bands of Himawari-8' sensor

TABLE III
Number of Images in Each Class in LSCIDMR and LSCIDMR-V2

Type	LSCIDMR		LSCIDMR-V2	
	Single	Multi	Single	Multi
Tropical Cyclone (TC)	3305	3305	3305	3305
Extratropical Cyclone (EC)	4984	4984	4984	4984
Frontal Surface (FS)	634	634	634	634
Westerly Jet (WJ)	628	628	628	628
Snow	7631	8700	7631	8700
Vegetation(Veg)	7831	47421	4036	47421
Desert	4518	59448	3906	59448
Ocean	4042	93756	1754	47421
Low Water Cloud	1774	99312	-	-
High Ice Cloud	5278	96033	-	-
Cirrus (Ci)	-	-	10289	69551
Cirrostratus (Cs)	-	-	4538	59246
Deep Convection (DC)	-	-	156	33335
Altocumulus (Ac)	-	-	2776	76985
Altostratus (As)	-	-	1724	53654
Nimbostratus (Ns)	-	-	215	22069
Cumulus (Cs)	-	-	3097	59274
Stratocumulus (Sc)	-	-	6176	62519
Stratus (St)	-	-	2	7860
LabelLess	63765	-	48539	-

The newly added labels of cloud category in LSCIDMR-V2 are shown in **bold**. And "-" means no such label in this corresponding dataset.

A. Implementation Details

We adopt four shallow CNNs as the feature extractor for four kinds of contents, and each CNN could be divided into three blocks. Each block consists of a convolutional layer, a Relu as activation function, and a max pooling for downsampling purposes. The first Conv layer has 64 feature maps with a kernel size of 7×7, the second Conv layer has 256 feature maps with a kernel size of 5×5, and the last contains 512 feature maps with a kernel size of 3×3. The size of features for each content we extracted is $D \times H \times W = 512 \times 5 \times 5$. The hidden size of the transformer block is set to 512 consistent with the depth of features. The number of layers of the transformer encoder is set to 2, and the number of heads is set to 4. We modified all competing methods' first conv layers with 16-channel-input ones so that they can take 16-channel images as input. Also, in the training process, we utilize Adam as our optimizer, and the weight decay is set to 0.0004. The learning rate would be reduced to 1/5 of the original learning rate when the mean average precision (mAP) on the valid set is not improving for three epochs, and the learning rate is initialized to 5×10^{-5}. The batch size is set to 64. Simple data augmentations, such as random rotation and random flip, are performed for better generalization ability. All experiments are conducted on an NVIDIA Quadro RTX A6000 GPU with 48-GB memory.

In terms of the partition of LSCIDMR-V2, every ten days, one day's data were selected as validation data, another day's data were selected as test data, and the rest of the data serve as the training set; hence, the validation ratio and test ratio are both fixed at 10.14%. This partitioning method prevents the space and time overlap between the training set and

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TABLE IV
OVERALL COMPARISON OF DIFFERENT METHODS ON LSCIDMR-V2

Method	mAP	subacc	CP	CR	CF1	OP	OR	OF1
ResNet-50	73.61	20.21	68.56	68.01	68.29	85.49	85.12	85.30
ResNet-101	73.50	19.31	68.83	67.10	67.96	86.54	84.40	85.46
ResNet-152	74.18	18.88	70.59	65.83	68.13	85.80	84.82	85.31
VGG-16	79.74	36.27	79.55	73.80	76.57	91.65	91.57	91.61
VGG-19	79.60	35.77	**82.88**	70.84	76.39	91.95	91.34	91.64
ADD-GCN(ResNet-101)	74.74	21.83	71.24	65.79	68.41	87.58	86.12	86.84
ADD-GCN(s-CNN)	75.04	23.94	71.61	67.09	69.27	88.42	89.41	88.92
C-Tran(ResNet-101)	78.62	28.20	73.38	73.60	73.49	87.70	91.59	89.60
C-Tran(s-CNN)	78.21	37.95	75.58	70.55	72.98	91.31	92.06	92.18
Q2L(TResNetL)	75.41	27.37	71.46	69.60	70.52	87.34	90.12	88.71
Q2L(s-CNN)	81.53	42.70	78.30	75.62	76.94	92.73	93.03	92.88
MS-GoGo	**82.40**	**44.35**	80.88	**75.63**	**78.17**	**93.21**	**93.43**	**93.32**

TABLE V
APS OF DIFFERENT METHODS ON EACH CATEGORY

Method	mAP	AP																
		TC	EC	FS	WJ	Snow	Ocean	Desert	Veg	Ci	Cs	DC	AC	AS	Ns	Cu	Sc	St
ResNet-50	73.61	35.24	48.16	8.45	5.98	86.79	99.73	94.18	97.65	94.30	96.67	90.00	94.79	92.21	84.45	95.02	95.94	31.83
ResNet-101	73.50	28.09	47.70	8.21	1.73	88.31	99.85	94.60	97.23	95.77	97.54	91.69	95.82	92.98	83.30	94.57	97.69	34.32
ResNet-152	74.18	44.81	46.18	16.45	2.55	86.65	99.74	94.36	97.60	95.51	96.83	91.41	94.30	91.07	81.17	92.28	96.27	33.84
VGG-16	79.74	54.92	46.05	18.02	**16.40**	92.24	99.96	96.96	98.74	97.98	98.74	95.67	98.30	95.80	91.27	96.38	98.82	59.32
VGG-19	79.60	52.47	43.56	**25.36**	10.37	91.78	99.95	97.04	98.80	98.27	98.64	95.98	98.35	95.83	91.13	97.28	98.98	59.44
ADD-GCN(ResNet-101)	74.74	49.85	47.74	7.49	7.85	85.70	99.84	95.40	97.75	95.16	96.61	89.46	95.51	92.43	84.19	94.93	98.04	32.71
ADD-GCN(s-CNN)	75.04	47.85	47.88	4.70	2.49	86.55	99.92	96.65	98.60	95.93	95.45	86.90	95.50	92.90	82.30	95.40	97.87	48.87
C-Tran(ResNet-101)	78.62	**77.55**	**62.06**	13.64	2.72	88.24	99.85	95.83	98.33	96.67	97.62	91.40	96.04	92.20	86.80	96.85	98.67	42.04
C-Tran(s-CNN)	78.21	38.98	52.29	5.97	2.14	91.55	99.94	97.21	98.93	98.41	98.89	96.49	98.58	97.08	91.76	98.15	99.25	63.88
Q2L(TResNetL)	75.41	47.52	45.55	9.75	2.84	89.60	99.93	96.31	98.31	96.78	97.89	93.10	96.19	92.37	83.72	96.51	98.58	37.10
Q2L(s-CNN)	81.53	69.49	54.80	16.35	4.36	93.77	99.96	97.78	99.13	98.72	**99.13**	**97.08**	98.72	97.37	**92.85**	98.34	**99.32**	68.77
MS-GoGo	**82.40**	72.82	54.93	25.22	4.34	**94.12**	**99.97**	**98.66**	**99.24**	**98.92**	99.10	96.85	**98.93**	**97.45**	92.58	**98.40**	99.26	**69.94**

validation/test set so that the model's performance and generalization ability could be shown more objectively. Because if the model simply memorizes the training sample and the space and time overlap between the training set and validation/test exists, its performance on validation/test will be superficially high with limited generalization ability. Furthermore, in order to balance the data loading efficiency and the training effect, the training set was randomly shuffled at the day level. In this way, we can load one day's data together and perform the patch cropping process in the RAM to alleviate the IO cost of the overlapped part.

B. Metrics

We report mAP, subset accuracy (subacc), precision [per-category precision (CP) and overall precision (OP)], recall [per-category recall (CR) and overall recall (OR)], and F1 (CF1 and OF1) scores to show the performance of methods. A higher number of these metrics mean better performance.

The mAP is the mean value of all categories' APs. AP summarizes a precision–recall curve as the weighted mean of precisions achieved at each threshold, with the increase in recall from the previous threshold used as the weight

$$\mathrm{AP} = \sum_n (R_n - R_{n-1}) P_n \tag{9}$$

where P_n and R_n are the precision and recall at the nth threshold, respectively. Subacc, the percentage of the sample that is predicted fully correct, is also presented. The number of fully correct samples is notated as M_{fc}, and as the number of all samples is notated as M, then subacc can be defined as

$$\mathrm{subacc} = \frac{M_{\mathrm{fc}}}{M}. \tag{10}$$

What is more, the number of images, which is predicted correctly for the ith category, the number of images predicted for the ith category, and the number of ground truth images for the ith category, is notated as M_c^i, M_p^i, and M_g^i, respectively. Then, other metrics, OP, CP

$$\mathrm{OP} = \frac{\sum_i M_c^i}{\sum_i M_p^i}, \quad \mathrm{CP} = \frac{1}{C}\sum_i \frac{M_c^i}{M_p^i} \tag{11}$$

overall recall (OR), CR

$$\mathrm{OR} = \frac{\sum_i M_c^i}{\sum_i M_g^i}, \quad \mathrm{CR} = \frac{1}{C}\sum_i \frac{M_c^i}{M_g^i} \tag{12}$$

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TABLE VI
COMPARISON OF Q2L AND MS-GOGO'S SELECTED LABELS

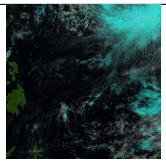
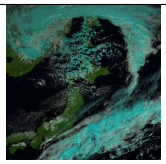
	Ground Truth	Q2L (s-CNN)	MS-GoGo
 20190810_10_11	TC, Veg, Desert, Ocean, Ci, Cs, Ac, As	Veg, Desert, Ocean, Ci, Cs, Ac	**TC**, Veg, Desert, Ocean, Ci, Cs, Ac, **As**
 20190929_02_13	EC, Veg, Desert, Ocean, Ci, Cs, DC, Ac, As, Cs, Sc	Veg, Desert, Ocean, Ci, Cs, DC, Ac, As, Cs,	**EC**, Veg, Desert, Ocean, Ci, Cs, DC, Ac, As, Cs, **Sc**

TABLE VII
EFFICIENCY COMPARISON OF DIFFERENT METHODS

	Model Size (MB)	Parameters (M)	Training Time (ms)	Inference Time (ms)
ResNet-50	94.65	23.58	4.44	2.02
ResNet-101	170.94	42.58	6.37	2.64
ResNet-152	233.80	58.22	8.50	3.32
VGG-16	537.36	134.34	2.96	2.88
VGG-19	558.60	139.65	3.39	3.15
ADD-GCN(ResNet-101)	192.22	47.88	6.49	2.81
ADD-GCN(s-CNN)	16.30	4.07	1.62	1.44
C-Tran(ResNet-101)	380.83	95.04	8.72	3.82
C-Tran(s-CNN)	14.05	3.51	1.80	1.69
Q2L(TResNetL)	980.11	244.38	21.96	6.94
Q2L(s-CNN)	20.49	28.70	2.32	2.23
MS-GoGo	22.99	5.73	2.87	2.56

The Training Time and Reference Time are the average training time and inference time on each sample with an NVIDIA Quadro RTX A6000 GPU.

and F1-measures (CF1 and OF1)

$$\mathrm{OF1} = \frac{2 \times \mathrm{OP} \times \mathrm{OR}}{\mathrm{OP} + \mathrm{OR}}, \quad \mathrm{CF1} = \frac{2 \times \mathrm{CP} \times \mathrm{CR}}{\mathrm{CP} + \mathrm{CR}} \tag{13}$$

are also presented to better demonstrate the performance of our model. Among them, the most important metrics are mAP, subacc, OF1, and CF1 since they are comprehensive metrics [26].

C. Overall Performance on LSCIDMR-V2

ADD-GCN [33], C-Tran [26], and Q2L [27] are the methods we compared with. ADD-GCN is a graph-based MILC method and the latter two methods are based on transformers. Besides these latest methods, popular baselines, such as VGG-16, VGG-19, ResNet-50, ResNet-101, and ResNet-152, are included in the comparison. The overall performance is listed in Table IV, and the performance by category is listed in Table V; the best results are shown in **bold**. As we mentioned in [19], shallow CNNs may perform better at RS image feature extraction; for a fair comparison, when compared with C-Tran, ADD-GCN, and Q2L, not only their default backbones are utilized, but we also switch on the shallow CNN (mark with "s-CNN" in parentheses in Table IV) we used in MS-GoGo.

Overally speaking, MS-GoGo outperforms other methods and achieves the SOTA performance across almost all metrics. As for the categorywise comparison, we can draw a conclusion from Table V that MS-GoGo achieves the best results in most categories compared to other methods. It should be noted that almost all models, including MS-GoGo, perform worse for classes TC, EC, FS, and WJ than other classes because of the relatively small amount of annotation of these labels.

Our method shows the most significant advantages over other methods on metrics subacc, which is the proportion of samples that are completely correctly predicted. We contribute this advantage to the introduction of prior knowledge of location and time information. The introduced information helps MS-GoGo to recognize the labels' co-occurrence pattern of a given location at a given time, which is valuable for the model to make precise predicts. For example, as shown in Table VI, for sample 20190810_10_11, it is difficult for Q2L (Q2L with s-CNN has the second highest subacc) to recognize the TC in the image since only the edge part of TC is included. However, because we introduced time and location information into MS-GoGo and the knowledge that TCs are common in the North Pacific in August can be learned by our model; combined with the information from the MS image, MS-GoGo can pick out the label TC with more confidence. It is worth noting that the emergence of Altostratus (As) in summer is highly related to TC, which is also recognized by MS-GoGo. Furthermore, Q2L is not able to recognize the EC and Sc in sample 20190929_02_13 (northern part of Japan is included) since only half of an EC is in the image. However, because of the prior knowledge that EC is likely to occur over northern Japan in September, label EC as well as Sc, which is common over coastal areas, in this sample, are correctly predicted by MS-GoGo. These examples demonstrate why there is a higher subacc of MS-GoGo.

The model size, the number of parameters, and training/inference time are reported in Table VII. It can be seen from Table VII that our proposed MS-GoGo is much lighter than ResNets, VGGs, as well as ADD-GCN (ResNet-101), C-Tran (ResNet-101), and Q2L (TResNetL). The model size, the number of parameters, and training/inference time of MS-GoGo are at the same level as ADD-GCN (s-CNN), C-Tran (s-CNN), and Q2L (s-CNN). Besides, Table IV shows that the proposed MS-GoGo gains better accuracy than all the compared methods.

D. Evaluation of Group Features Extractor

We compare our group extractor with the other two feature extracting methods (shown in Fig. 5)—dealing with all channels together with a single CNN and treating each channel separately with the channel number of CNNs. Table VIII shows our experimental result. We can see that in the same circumstance, the group feature extractor is superior to other

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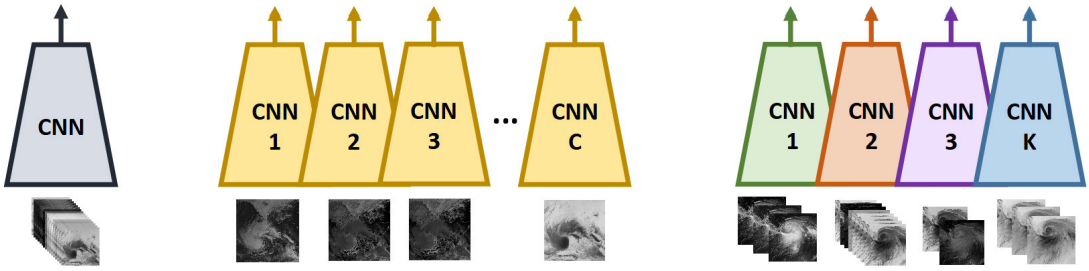


Fig. 5. Different ways to extract multichannel RS images. (a) Single extractor treats the RS image as a whole and is only able to extract the common feature of all channels. (b) Separate extractor copes with each channel separately but cannot take full advantage of the relationships between channels. (c) With a group extractor, feature extraction is carried out according to the reflected contents, and prior knowledge of diversified contents carried by the MS image is fully utilized. "c" denotes the total number of channels of MS images, and "k" denotes the total number of groups that can be divided.

TABLE VIII
COMPARISON OF DIFFERENT FEATURE EXTRACTORS

	mAP	subacc	OF1	CF1
Separate-3c	78.54	35.07	72.59	91.20
Single-3c	79.33	38.47	74.81	92.11
Separate-16c	81.27	40.06	**78.35**	92.43
Single-16c	81.49	41.71	77.63	92.88
Group-16c	**81.91**	**42.87**	77.75	**92.99**

"3c" denotes only 3 channels (3rd,4th,5th channel of HIMAWARI-8 MS data) are used and "16c" denotes all 16 channels are used.

TABLE IX
RESULTS OF DIFFERENT QUERIES WITH IMAGE

	mAP	subacc	OF1	CF1
with image (naked)	81.91	42.87	77.75	92.99
with image (Geo-Queries)	**82.40**	**44.35**	**78.17**	**93.32**

TABLE X
RESULTS OF DIFFERENT QUERIES WITHOUT IMAGE

	mAP	subacc	OF1	CF1
random value	37.37	0.01	41.77	44.45
without image (naked)	37.39	0.84	28.35	65.19
without image (Geo-Queries)	**70.91**	**21.47**	**65.69**	**86.88**

random value: obtained by the randomly initialized model without any training

two ways, and the separate extractor is the worst way to choose. Dealing with all channels together with a single CNN is a convenient way to extract RS image features, but due to its limitation of utilizing diversified contents, the performances of it are worse than group feature extractor. Taking advantage of the diversity of RS image contents, one way is taking down the original MS image apart and treating each channel separately. However, its performance still cannot beat the group feature extractor and is even worse than the "together way" since it is a difficult task for the model to find out the correlation of channels without any other prior knowledge.

We attribute group feature extractor's success to manually introduce priori knowledge into the model by grouping channels, which reflects similar contents in group and extracts the features of these groups separately. In other words, a group feature extractor can process similar contents together while differentiating between distinct contents. These diversified yet complementary features then interact with each other via the transformer encoder block, and the whole picture of an MS image is illustrated by the model. Experiments show that such an attempt has yielded promising results.

To better demonstrate our perspective that the rich contents contained by MS image would benefit the performance of the model, we made a comparison between the way utilizing all 16 channels of Himawari-8 and the way using just three channels. The three-channel data, consistent with original LSCIDMR [21], contains the 3rd, 4th, and 5th channel of Himawari-8. For three-channel data, we carry out three-channel together way (single extractor) and three-channel separate way (only three channels cannot be grouped) to compare with 16-channel data. We can draw the conclusion from Table VIII that the MS information can support a better performance than only three channels.

E. Evaluation of Geo-Queries

To evaluate the value of the proposed geo-queries, we observe how the performance changes when we add month embedding and location embedding to form geo-queries. Also, we call the queries naked queries if they do not contain location or month embeddings. To figure out how much information could be learned exactly by geo-queries, we also conduct experiments without the input of images.

We can draw conclusions from Table IX that, when we replace the original naked queries with geo-queries, the performance of our model is significantly improved. It is worth noting that the addition of location and time information would greatly improve the subacc, which indicates that such information is valuable for deep-learning models to learn more precise knowledge to distinguish the characteristics and differences among samples.

Apart from the improvement of final metrics, we find that the addition of month and location embedding will accelerate the convergence of the model. Fig. 6 shows the convergence of models during training. The model with geo-queries has more rapid convergence compared to the one with naked queries.

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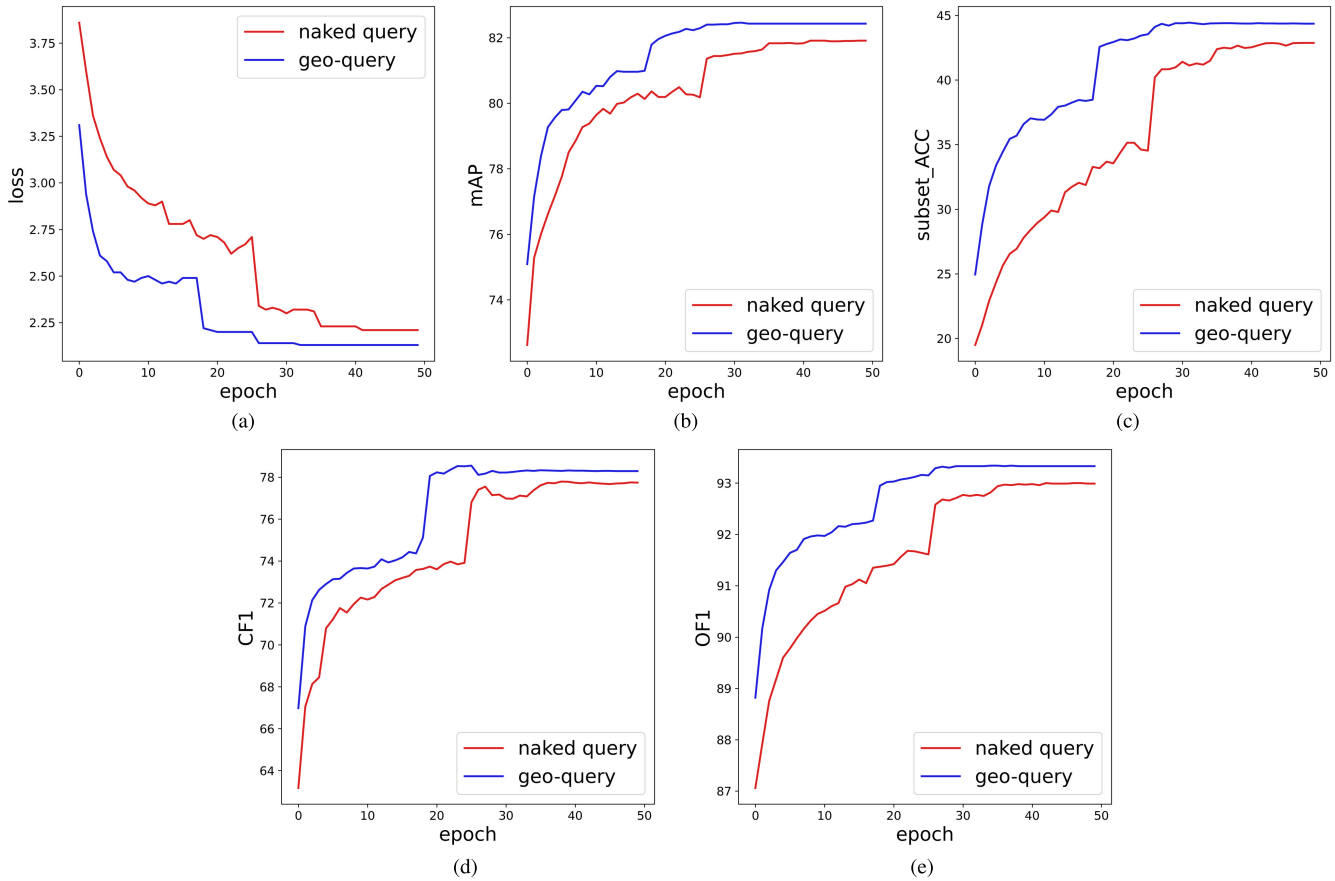


Fig. 6. Converge under naked queries and geo-queries. (a)–(e) Change of loss, mAP, sub-acc, CF1, and OF1, respectively, on test set with the increase of epochs. The convergence of model with geo-queries would be faster. (a) Loss change. (b) mAP change. (c) Sub-acc change. (d) CF1 change. (e) OF1 change.

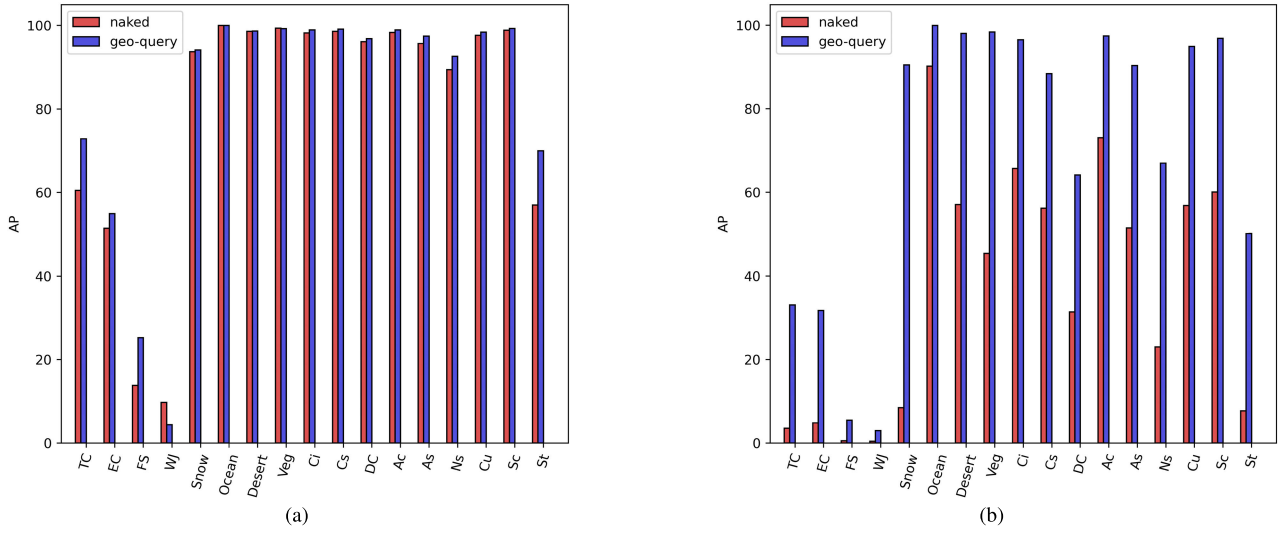


Fig. 7. MPs for every class. (a) MPs with group feature extractor and (b) MPs without image signals to illustrate how much information geo-queries can learn. With the addition of month and location embeddings, the performance of the model shows an obvious improvement.

This phenomenon is also a powerful argument that the added information is helpful for our model to recognize different components in MS images.

Let us analyze the effect of geo-queries on each category to better illustrate the value of the addition of time and location information. Fig. 7(a) shows the performances on each category with different queries. We can see except for WJ, and the addition of location and month information, to a certain extent, improves the identification accuracy of each category. Also, we think that the performance degradation in category WJ may be due to the limitation of the sample amount of this category according to Table III.

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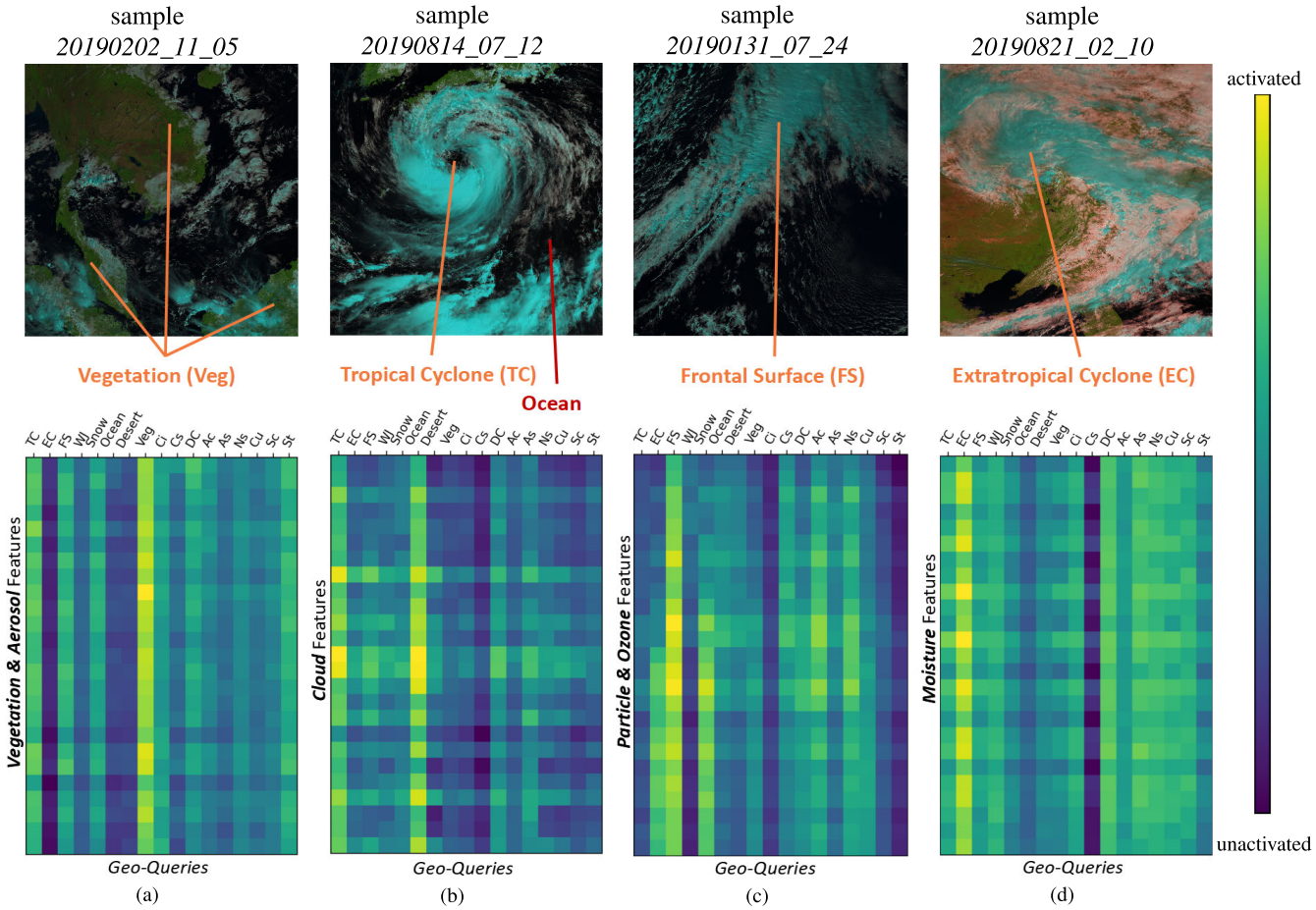


Fig. 8. Visualization of the attention maps in the transformer encoder block between specific features and geo-queries. The selected sample patches are shown in the first row and the corresponding attention maps are shown in the second row. The vertical axis of attention maps represents the tokens of specific features captured by the group feature extractor, and the horizontal axis represents geo-queries. (a) Sample 20190202_11_05 includes the Indo-China peninsula covered by vegetation (Veg), while the attention map below tells that the geo-query of veg is activated by its vegetation and aerosol features. (b) Sample 20190814_07_12 shows a TC surrounded by oceans; accordingly, the geo-queries of TC and ocean are activated by its cloud features. (c) Sample 20190131_07_24 shows an FS and the geo-query of FS is activated by its particle and ozone features. (d) Sample 20190821_02_10 shows an EC, while the geo-query of EC is activated by its moisture features.

Our model achieves promising results with the combined efforts of both MS image information and geoinformation (location and time). We have already seen how our model will perform without geoinformation according to Table IX. However, how about we get rid of the image information and let the geoinformation works alone? To figure out this question, we conduct an experiment that removes all image signals to observe how our models perform under such a condition. We reported the results of different queries without image in Table X, where the first row "random value" presents the baseline results obtained by the randomly initialized model without any training. From Table X, we can see that the addition of month embeddings and location embedding will greatly improve the performance of the model. Also, we are hugely surprised to find out that without any image information, and only depending on our geo-queries, our model could achieve such a great performance. This demonstrated that there is a strong correlation between a given location and distributions of particular weather systems, cloud types, and land covers at a given time. Also, through the training process, the association pattern between categories is successfully learned by our model, so time and location could be utilized to identify different categories independently with relatively high accuracy. However, this information cannot replace images but can be positive complements to them. In Fig. 7(b), we can see that the introduction of location and time information is very valuable for all categories. Also, another interesting fact is that, when we replace naked queries with geo-queries, the MP of snow, desert, and veg, which are closely related to location and seasonal information, obtains a huge growth. This proves the opinion that our location embeddings and month embeddings indeed have the ability to learn an effective representation of location and time information, which is valuable for component recognition.

F. Visualization of Data Fusion and Content Querying

To further show the validity of the proposed MS-GoGo, the attention maps of the transformer block in the data fusion and content querying module are visualized, and Fig. 8 shows some examples. As mentioned in Section III-D, different features captured by the group feature extractor are fed into an N-layer transformer block, and geo-queries are responsible

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for probing the semantic-similar ingredients of these features. In other words, if some semantic ingredients are included in a specific feature, corresponding geo-queries would be activated with the help of the self-attention mechanism, and the corresponding region of the attention map would be lit up.

Specifically, as shown in Fig. 8, some representative samples are selected and their corresponding attention map in the transformer blocks between tokens of specific features generated by group feature extractor and geo-queries are presented below the patches. As shown in Fig. 8(a), barely obscured by clouds, sample 20190202_11_05 shows the Indo-China Peninsula covered by vegetation (veg); accordingly, the attention map below displays that the geo-query of veg is activated by vegetation and aerosol features. Sample 20190814_07_12 shown in Fig. 8(b) includes a typical TC, and the corresponding region of the attention map of TC geo-query is lit up by the cloud feature; these features also noticed the occurrence of the surrounding ocean, and thus, the geo-query of ocean is also activated. Fig. 8(c) shows sample 20190131_07_24, which includes an FS, whose occurrence is captured by the particle and ozone features, and the geo-query of FS is activated by these features. Sample 20190821_02_10 in Fig. 8(d) shows an EC, and its occurrence is captured by the corresponding moisture features, and these features activate the geo-query of EC as shown in the attention map next.

VI. Conclusion

In this article, a novel deep-learning method of RSMLIC named MS-GoGo and a new RS multilabel image dataset called LSCIDMR-V2 are proposed. Via group feature extractor and geo-queries, the diversified content and accompanying geographic information of MS images are effectively utilized by the proposed MS-GoGo. Our approach achieves the state-of-the-art performance according to our experimental results. We also demonstrate the contribution of the components in our model through a series of ablation experiments, which shows that: 1) the MS images' contents can be utilized more effectively and the accuracy of the model can be improved by grouping feature extraction of channels and 2) by adding month and location embedding to form geo-queries, the time and location information are successfully introduced in our deep-learning method, which is beneficial for the performance of our model. The proposed LSCIDMR-V2 can provide data support for the application of the deep-learning method in the field of meteorological research and the evaluation of other RSMLIC methods. For future works, we plan to improve the granularity of time and location embedding to adapt to more general RS image recognition tasks. Semisupervised and unsupervised methods should also be considered since there is still a data gap between us and a better model.

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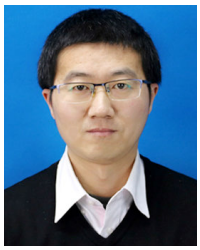
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